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Fluidic connector and method for assembling it

Abstract

A fluidic connector includes a female tubular fitting, a male tubular fitting, and a generally U-shaped metal pin mounted astride the female fitting. The pin is axially movable on the female fitting from an inoperative position in which branches of the pin are held spaced apart by the female fitting, to an operative position in which portions of the branches are located at the level of at least one slit of the female fitting and are able to pass through the slit by elastic return. The male fitting includes a part configured to be engaged in the female fitting and to cooperate directly with the branches of the pin move the male fitting from its inoperative position to its operative position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority from Patent Application No. 2211282 filed on Oct. 28, 2022 in France. The entire contents of the priority application is incorporated herein by reference.

TECHNICAL FIELD OF THE INVENTION

(2) The present invention relates to a fluidic connector and its assembly method.

PRIOR ART

(3) The technical background comprises the documents U.S. Pat. No. 11,168,819B2, AT-B1-512 397 and U.S. Pat. No. 7,438,328B2.

- (4) A fluidic connector is a device that allows to connect two tubular fittings in a fluidic circuit to allow the circulation of fluid through the fittings and into the circuit. The two fittings are generally male and female fittings and therefore these two fittings are connected by a male-female fitting.
- (5) To prevent accidental disconnection of the two fittings of the connector, the connector has been known to be equipped with a lock. The lock is carried by one of the fittings, for example the female fitting. When the male and female fittings are engaged, the lock can cooperate with the male fitting to retain it axially in the female fitting. In this case, the axial direction is the direction in which the fittings engage with each other.
- (6) In the context of the present invention, the lock is a metal pin which is generally U-shaped and is mounted astride the female fitting. The pin comprises two lateral branches which pass through slits in the female fitting and can engage in a gorge in the male fitting.
- (7) The pin is elastically deformable by moving its branches apart and close together. When the branches are moved close together, they engage in the gorge of the male fitting and block it axially with respect to the female fitting. When the branches are spaced apart, they are extracted from the gorge of the male fitting, which can be removed from the female fitting by axial translation.
- (8) The passage from the spaced apart position to the close together position of the branches of the pin is preferably carried out automatically by elastic return of these branches. In practice, when the male fitting is inserted into the female fitting, the male fitting rests on the branches of the pin and pushes them radially outwards to space them apart from each other. The radial direction is expressed here in relation to the aforementioned axis. This insertion is continued until the gorge of the female fitting and the pin are located in the same plane perpendicular to the axis. The branches then move close together by elastic return and are housed into the gorge of the female fitting.
- (9) In the current technique, it is a frustoconical part of the male fitting that cooperates with the branches of the pin when it is inserted into the female fitting. During the insertion, the branches move apart from each other as they slide over this frustoconical part, from the axial end of the smaller diameter of this frustoconical part to its axial end of larger diameter.
- (10) The axial retention capacity of the male fitting in the female fitting depends in particular on the force required to move the branches apart. It is therefore preferable for this force to be relatively high, to avoid the branches moving apart unintentionally. To generate this force, it is necessary to force-fit the male fitting into the female fitting so that, as mentioned above, the male fitting cooperates with the branches and exerts the force required to move them apart.
- (11) With the current technology, the male fitting is forced-fitted into the female fitting manually, which can be difficult for an assembly operator to do.
- (12) The present invention is an improvement on the current technology, providing a simple, effective and economical solution to this problem.

SUMMARY OF THE INVENTION

- (13) The invention relates to a fluidic connector, comprising: a female tubular fitting with a first axis, a tubular male fitting having a second axis and configured to be engaged coaxially with the female fitting, and a generally U-shaped metal pin mounted astride the female fitting, this pin comprising two lateral branches, portions of which are configured to pass through at least one slit in the female fitting and to be engaged in a gorge in the male fitting to block it axially relative to the female fitting, the pin being elastically deformable by moving its branches apart and close together and being able to adopt several configurations, in particular between a close together configuration in which said portions are at a distance $D1$ from each other and are housed in the gorge, and a spaced apart configuration in which said portions are at a distance $D2$ from each other and allow the male fitting to be removed from the female fitting, $D2$ being greater than $D1$, characterised in that the pin is axially movable on the female fitting from an inoperative position in which its branches are held spaced apart by the female fitting so that said portions are at a distance $D3$ from one another, $D3$ being between $D1$ and $D2$, to an operative position in which said portions are located at the level of said at least one slit and are able to pass through it by elastic return, and

in that the male fitting comprises a part configured to be engaged in the female fitting and comprising an external diameter which is greater than $D3$, so that, when the male fitting is engaged in the female fitting, this part cooperates directly with said portions to move the pin from its inoperative position to its operative position.

(14) In the present application, “position” means a position of the pin and in particular two positions are defined for the pin, namely an inoperative position and an operative position. By “configuration” we mean the state of the branches of the pin, more or less spaced apart or close together. The branches can be in a position (extreme—corresponding to the distance $D2$) that is spaced apart enough for the male fitting to disengage from the female fitting. The branches can be in a position (extreme—corresponding to the distance $D1$) close together to allow the axial locking of the male fitting in the female fitting. The branches can also be in an (intermediate—corresponding to the distance $D3$) spaced apart position to cooperate with the male fitting when it is inserted into the female fitting.

(15) In the prior technology, the branches of the pin can adopt essentially two extreme “stable” configurations, namely a spaced apart configuration and a close together configuration. In the present invention, the pin can adopt an intermediate “stable” configuration in which its branches are spaced apart but less than in the extreme spaced apart configuration.

(16) The connector according to the invention is intended to be delivered with its pin whose branches are in this intermediate configuration. As a result, when an operator engages the male fitting in the female fitting, the force required to space the branches apart is lower than in the prior technique, since the branches are already slightly spaced apart. This is because the force required corresponds to the spacing of the branches from this intermediate configuration to the extreme spaced apart configuration (and not from the extreme close together configuration to the extreme spaced apart configuration). This makes it easier for the operator to mount the fittings.

(17) It is therefore understood that the pin is mounted on the female fitting in a pre-stressed manner so that its branches are already spaced apart from each other in the inoperative position. It is also understood that the pin is moved axially on the female fitting to pass from its inoperative position to its operative position in which its branches can be housed in the gorge of the male fitting by elastic return.

(18) In the present invention, it is the male fitting that cooperates with the pin and moves it over the female fitting. The force with which the male fitting engages the female fitting therefore firstly moves the pin on the female fitting and secondly moves the branches of the pin further apart. This force is still less than the force required in the prior technique to move the branches apart between their end positions.

(19) In this application, “upstream” and “downstream” are also defined as locations relative to the direction of engagement of the male fitting in the female fitting. The male fitting thus has an upstream free end which is engaged in a downstream free end of the female fitting.

(20) The connector according to the invention may comprise one or more of the following characteristics, taken in isolation from each other, or in combination with each other: in said operative position, the pin extends mainly in a plane perpendicular to said first axis, and in said inoperative position, the pin extends mainly inclined with respect to said first axis; in said inoperative position, the pin is inclined so that free ends of the branches are located downstream relative to opposite ends of the branches, with respect to a direction of engagement of the male fitting in the female fitting; the female fitting comprises two slits which are substantially diametrically opposed with respect to said first axis and which each have an elongated shape in the circumferential direction about this first axis, each of these slits being intended to be passed through by one of said portions; the female fitting further comprises two notches which are substantially diametrically opposed with respect to said first axis and which communicate respectively with said slits, each of the notches having a lateral wall which is inclined with respect to a plane perpendicular to said first axis and on which one of said portions is adapted to engage by

axial abutment when the pin is in its inoperative position; each of the notches is generally V-shaped or wedge-shaped and comprises a thin end and a widened end located on the side of a free end of the corresponding branch; each of the notches comprises a first part which opens into the interior of the female fitting, and a second part which defines a step, this step being configured to cooperate by radial abutment with a free end of the corresponding branch, in order to hold the pin in its inoperative position; said step comprises a projecting wedge on the side of the corresponding slit, this projecting wedge comprising a lug for retaining the pin in its inoperative position; said step is connected, on the side opposite the lug, to a stop for the pin; the steps of the female fitting comprise parallel sliding surfaces of the free ends of the branches; in its operative position, the pin is able to move in a plane perpendicular to said first axis, and in its inoperative position, the pin is able to move in a different plane, which is preferably inclined with respect to said first axis; said part of the male fitting is a frustoconical part which comprises an upstream end of smaller diameter and a downstream end of larger diameter, this part cooperating by bearing with said portions in areas which are closer to its downstream end than to its upstream end; the female fitting comprises recesses which are substantially diametrically opposed with respect to said first axis, and which are configured to accommodate bent free ends of said branches when the pin is moved from its operative position to a disassembly position in which the pin has said spaced apart configuration;

(21) The present invention also relates to a fluid circuit for a motor vehicle, comprising at least one connector as described above.

(22) The present invention also relates to a method for assembling a fluidic connector as described above, comprising the steps of: mounting the pin on the female fitting, with the pin in its inoperative position, and engaging the male fitting in the female fitting, this engagement causing the pin to move from its inoperative position to its operative position in which said portions are located at the level of said at least one slit and are housed in the gorge of the male fitting by elastic return.

Description

BRIEF DESCRIPTION OF THE FIGURES

(1) Further characteristics and advantages will be apparent from the following description of a non-limiting embodiment of the invention with reference to the appended drawings in which:

(2) FIG. 1 is a schematic perspective view of a female fitting and a pin for a connector according to the invention;

(3) FIG. 2 is a schematic perspective view of a male fitting for a connector according to the invention;

(4) FIG. 3 is a schematic front view of the male fitting shown in FIG. 2;

(5) FIGS. 4a and 4b are schematic perspective views of two parts of the female fitting of FIG. 1;

(6) FIG. 5 is a schematic axial sectional view of the female fitting shown in FIG. 1;

(7) FIGS. 6a-6b are schematic perspective views of one of the parts of the female fitting shown in FIG. 1;

(8) FIG. 7 is a schematic perspective view of the pin shown in FIG. 1;

(9) FIGS. 8a and 8b are schematic side and front views, respectively, of the female fitting and of the pin of FIG. 1, and illustrate the pin in an inoperative position;

(10) FIG. 9 is a schematic side view of the connector according to the invention, which comprises the female fitting and the pin of FIG. 1 as well as the male fitting of FIG. 2, and illustrates the pin in an operative position and its branches in a close together configuration;

(11) FIG. 10 is a schematic axial sectional view of the connector shown in FIG. 9;

(12) FIG. 11 is a schematic front view of the connector in FIG. 9 without the male fitting; and

(13) FIGS. **12a** and **12b** are schematic side and front views respectively of the female fitting and of the pin of FIG. **1**, and illustrate the pin in a position in which its branches are in a spaced apart configuration.

DETAILED DESCRIPTION OF THE INVENTION

(14) FIGS. **1** to **12b** illustrate an embodiment of a fluidic connector **10** according to the invention.

(15) The connector **10** is shown complete in FIGS. **9** and **10**.

(16) The connector **10** essentially comprises three elements: a male fitting **12**, a female fitting **14** and a pin **16**.

(17) The male fitting **12** is shown on its own in FIGS. **2** and **3**. The pin **16** is shown on its own in FIG. **7**. The female fitting **14** equipped with the pin **16** can be seen in FIGS. **1**, **4a-4b**, **5**, **6a-6b**, **8a-8b** and **12a-12b**.

(18) We are going to describe the elements of the connector **10** one after the other.

(19) The male fitting **12** (FIGS. **2** and **3**) is generally tubular in shape and comprises a main axis Y corresponding to the axis of elongation of its internal passage. Although the male fitting **12** has a straight elongated shape in the drawings, it could alternatively have a bent shape.

(20) The male fitting **12** comprises two axial stretches **12a**, **12b**. The first stretch **12a** is an engagement stretch designed to be engaged in the female fitting **14**. The second stretch **12b** is a connection stretch designed to be connected to a pipe or a conduit of a fluid circuit.

(21) The stretch **12a** comprises three successive parts, namely a cylindrical part **12a1**, a frustoconical part **12a2** and another cylindrical part **12a3**. The part **12a1** has a diameter H1 and the part **12a3** has a diameter H2 which is greater than H1. The part **12a2** extends from the part **12a1** to the part **12a3** and therefore has a diameter that varies from H1 to H2. The part **12a3** is located on the same side as the stretch **12a**, and the part **12a1** is located on the opposite side to the stretch **12a** and forms a free end of the male fitting **12**.

(22) The male fitting **12** comprises an annular gorge **22** at its external periphery. In the example shown, this gorge **22** is located in the part **12a3** and close to the part **12a2** (or at the intersection between the parts **12a2** and **12a3**).

(23) The gorge **22** has an axial dimension L. The gorge **22** has an internal diameter H3 and an external diameter H2 (FIG. **10**). In the example shown, H3 is greater than or equal to H1.

(24) The male fitting **12** may also comprise at least one projecting or hollow member **24**, for example on its part **12a3**, which is configured to cooperate by interlocking with a complementary member **26** of the female fitting **14**. In the example shown, there are two members **24**, diametrically opposed. They project from the part **12a3** and have a generally elongated shape along the axis Y. The cooperation of the members **24**, **26** may have an indexing and/or translational guiding function for the male fitting **12** in the female fitting **14**.

(25) The fitting **12** can be made of plastic, metal or a composite material. It can be integrally made.

(26) The pin **16** (FIG. **7**) is generally U-shaped and astride the female fitting **14** as shown in FIG. **1**. The pin **16** comprises two lateral branches **16a**, **16b** which are connected by an intermediate bridge **16c**.

(27) In the example shown, the branches **16a**, **16b** are essentially straight but could alternatively be curved. The free ends **28** of the branches **16a**, **16b** opposite the bridge **16c** are bent or curved, for example at 90°.

(28) The bridge **16c** can be of any shape, for example curved. Preferably, it comprises a median part **30** to make it easier for an operator to grip the pin **16**, either directly by hand or using a tool.

(29) The pin **16** is elastically deformable by moving its branches **16a**, **16b** apart and close together. The pin **16** can adopt several configurations, in particular between a close together (extreme) configuration visible in FIGS. **9** to **11**, and a spaced apart (extreme) configuration visible in FIGS. **10**, **12a** and **12b**.

(30) The branches **16a**, **16b** have portions **31** configured to pass through at least one slit **32** in the female fitting **14** and to engage in the gorge **22** in the male fitting **12** to block it axially with respect

to the female fitting **14**.

(31) When the branches **16a**, **16b** are in the (extreme) close together configuration visible in FIGS. **9** to **11**, the portions **31** are at a distance **D1** from each other (FIGS. **10** and **11**). In this configuration, the branches **16a**, **16b** and their portions **31** are not necessarily parallel and the distance **D1** is the smallest distance measured between the branches **16a**, **16b** inside the female fitting **14** and in a plane **P1** passing through the axis **X** of the female fitting **14** (FIG. **11**). In this configuration and as illustrated in FIG. **10**, the portions **31** are housed in the gorge **22** and it is therefore understood that the distance **D1** is close to or equal to **H3**.

(32) When the branches **16a**, **16b** are in the (extreme) spaced apart configuration shown in dotted lines in FIG. **10**, the portions **31** are at a distance **D2** from each other. In this configuration, the branches **16a**, **16b** and their portions **31** are not necessarily parallel and the distance **D2** is the smallest distance measured between the branches **16a**, **16b** in a plane **P1** passing through the axis **X** of the female fitting **14**. In the example shown, **D2** is close to or equal to **H2**. **D2** is therefore greater than **D1**. In this configuration, the branches **12a**, **12b** and their portions **31** are no longer housed in the gorge **22** of the male fitting **12**, which can be removed by pulling the female fitting **14** axially.

(33) According to the invention, the pin **16** can adopt an intermediate spaced apart configuration, as shown in FIGS. **8a** and **8b**. In this configuration, the portions **31** are at a distance **D3** from each other (FIG. **8b**). The branches **16a**, **16b** and their portions **31** are not necessarily parallel and the distance **D3** is the smallest distance measured between the branches **16a**, **16b** inside the female fitting **14** and in a plane **P1** passing through the axis **X** of the female fitting **14**.

(34) The distance **D3** lies between **D1** and **D2** and therefore between **H1** and **H2**. It is therefore understood that the frustoconical part **12a2** of the male fitting **12** is able to cooperate with the portions **31** of the pin **16** when its branches **16a**, **16b** are in this configuration. The cooperation of the male fitting **12** with the pin **16** will be described in detail after the female fitting **14** has been described in detail.

(35) When the branches **16a**, **16b** are in the (extreme) spaced apart configuration visible in FIGS. **12a** and **12b**, the portions **31** are at a distance **D4** from each other. In this configuration, the branches **16a**, **16b** and their portions **31** are not necessarily parallel and the distance **D4** is the greatest distance measured between the branches **16a**, **16b** in a plane **P1** passing through the axis **X** of the female fitting **14** (FIG. **12b**). In the example shown, **D4** is greater than **D2**. The pin **16** can be held in this configuration by its branches **16a**, **16b** and in particular its ends **28** cooperating with the female fitting **14**.

(36) The pin **16** is made of metal.

(37) The female fitting **14** (FIGS. **4a** and **4b**) is generally tubular in shape and has a main axis **X** corresponding to the axis of elongation of its internal passage. Although the female fitting **14** has a straight elongated shape in the drawings, it could alternatively have a bent shape. When the male fitting **12** engages in the female fitting **14**, the axes **X** and **Y** are coincident or aligned.

(38) The female fitting **14** comprises two parts **14a**, **14b** in the example shown, which are separated from each other in FIGS. **4a** and **4b**. Alternatively, the female fitting **14** could be formed from a single piece.

(39) The two parts **14a**, **14b** are designed to be nested into each other. The part **14a** in FIG. **4b** is an engagement part, a first stretch **14a1** of which is designed to receive the male fitting **12** and a second stretch **14a2** of which is designed to receive the other part **14b**.

(40) The part **14b** of FIG. **4a** is a connection part, a first stretch **14b1** of which is intended to be engaged in the stretch **14a2** and a second stretch **14b2** of which is intended to be connected to a pipe or a conduit of a fluid circuit. This second stretch **14b2** may, for example, comprise an external thread or anchoring ribs **35**, although this aspect is not limitative.

(41) In the example shown, which is not limitative, the part **14b** and in particular its stretch **14b1** comprises tabs **36** distributed around the axis **X** and configured to cooperate by elastic snap-fitting

with complementary means **37** of the part **14a** and in particular of the stretch **14a2** of the latter. This cooperation and the mounted position of the parts **14a**, **14b** in each other can be seen in FIG. 5.

(42) The female fitting **14** comprises an internal passage **34** comprising three parts **34a**, **34b** and **34c** designed to receive the male fitting **12**. The part **34a** is a part with a larger diameter **R1** which is slightly larger than the diameter **H2** so that this part **34a** can accommodate the part **12a3** of the male fitting **12** as shown in FIG. 10. This part **34a** is defined by the stretch **14a1** in the example shown. The part **34c** is a part with a smaller diameter **R2** which is slightly larger than the diameter **H1** so that this part **34c** can accommodate the part **12a1** of the male fitting as shown in FIG. 10. Finally, the part **34b** is a frustoconical part located between the parts **34a** and **34c** and has a diameter varying from **R1** to **R2**. This part **34a** is designed to accommodate the part **12a2** of the male fitting **12** as shown in FIG. 10.

(43) At its free end, the female fitting **14** and in particular its part **14a** comprises the aforementioned members **26**, which are in this case two diametrically opposed notches. This free end of the female fitting **14** defines the orifice **O** for inserting the male fitting **12**.

(44) The female fitting **14**, and in particular its part **14a**, comprises two diametrically opposed slits **32** through which the branches **16a**, **16b** of the pin **16** are intended to pass.

(45) Each slit **32** has an elongated shape in the circumferential direction around the axis **X**. For example, each slit **32** has a circumferential extent greater than or equal to 30° around the axis **X**.

(46) The female fitting **14**, and in particular its part **14a**, also comprises two diametrically opposed notches **38** which communicate respectively with the slits **32**. The notches **38** are formed downstream of the slits **32**.

(47) In the example shown, each notch **38** is generally V-shaped or wedge-shaped and comprises a thin end **38a** and a widened end **38b**. The thin end **38a** of the notch is located at the level of a free end of the corresponding slit **32** and communicates with this free end. The notch **38** also has an elongated shape in the circumferential direction and may have a circumferential extent similar to that of the slit **32**. The widened end **38b** of each notch **38** is located on the side of the other free end of the slit **32**.

(48) On the side opposite the orifice **O**, each slit **32** has a lateral wall **32a** which lies in a plane perpendicular to the axis **X**. The function of this wall **32a** is to guide the pin **16** during its deformation and it may extend beyond the slit **32**. In the example shown, the walls **32a** of the slits **32** are extended by first circumferential rims **40** provided on the female fitting **14** at its external periphery. These rims **40** are located upstream of the slits **32** and on an upper part of the fitting **14**. Each of the notches **38** has a lateral wall **38c** which is inclined with respect to a plane perpendicular to the axis **X**. The female fitting **14** comprises second circumferential rims **42** at its external periphery and at its upper end, which are axially spaced apart from the rims **40**, and which form extensions **38c'** of the lateral walls **38c** of the notches **38**. These extensions **38c'** may also be inclined with respect to a plane perpendicular to the axis **X**. The rims **42** are located downstream of the slits **32**. The rims **42** are substantially axially aligned with the rims **40**.

(49) The walls **39** of the notches **38** are designed to cooperate by axial support with the pin and in particular its branches **16a**, **16b**. The walls **32a** and the extensions **39'** are designed to cooperate with the branches **16a**, **16b** to guide them during their deformation.

(50) Each of the notches **38** comprises a first (upper) part **38c** which opens into the interior of the female fitting **14**, and a second (lower) part which defines a step **44**. The drawings, and FIG. 5 in particular, show that the slit **32** and the associated part **38c** of the notch **38** form a "P".

(51) The step **44** is configured to cooperate by radial support with a free end **28** of the corresponding branch **16a**, **16b**, as illustrated in FIG. 1 in particular.

(52) The step **44** comprises a projecting wedge **44a** on the side of the corresponding slit **32**. This projecting wedge **44a** preferably comprises a lug **46** for retaining this free end **28**. On the side opposite the projecting wedge **44a** or the lug **46**, the step **44** is connected to a stop **48** for the pin

16.

(53) Preferably, the steps **40** comprise outwardly oriented surfaces **40b** which are parallel and which are configured to allow the free ends **28** of the branches **16a**, **16b** to slide.

(54) Upstream of each slit **32**, the female fitting **14** comprises an outwardly oriented cylindrical surface portion **52**. The surface portions **52** are substantially axially aligned with the steps **44** and form radial bearing surfaces for the free ends **28** of the branches **16a**, **16b** and circumferential sliding surfaces for these free ends **28**.

(55) Each portion **52** extends around the axis X from an upper recess **50**, which is substantially axially aligned with the wedge **44a** of the corresponding step **44**, and a lower stop **54**.

(56) The recesses **50** are substantially diametrically opposed with respect to the axis X and are configured to accommodate the free ends **28** of the branches **16a**, **16b** in the spaced apart position of the pin **16**. These recesses **50** have an axial orientation and open out at the level of the walls **32a** of the slits **32**.

(57) The stops **54** are located just below the free ends of the notches **32** opposite the rims **40**, **42**.

(58) As mentioned above, the branches **16a**, **16b** of the pin **16** can adopt several configurations. The pin **16** can also adopt two distinct positions on the female fitting **14**, namely an inoperative position and an operative position.

(59) The inoperative and operative positions are axially spaced apart, which means that the pin **16** is moved axially on the female fitting **14** from its inoperative position to its operative position, and vice versa.

(60) In the inoperative position illustrated in FIGS. **1**, **8a** and **8b**, the pin **16** is inclined with respect to a plane perpendicular to the axis X and is therefore located in a plane P2 inclined with respect to the axis X. In this position, the free ends **28** of the branches **16a**, **16b** are located downstream with respect to the bridge **16c** and bear radially against the steps **40** and in particular their surfaces **40b**. The branches **16a**, **16b** and in particular their portions **31** are axially supported on the walls **39** of the notches **38**. In addition, the free ends **28** of the branches **16a**, **16b** can cooperate with the lugs **46** on the one hand and the stops **48** on the other hand to be held on the steps **40**. The ends of the branches **16a**, **16b** located on the side of the bridge **16c** can cooperate by axial abutment with the walls **32a** and the extensions **39'** of the rims **40**, **42** in order to prevent an accidental axial displacement of the pin **16** on the female fitting **14**.

(61) It is in this inoperative position in FIGS. **1** and **8a-8b** that the branches **16a**, **16b** of the pin **16** have the spaced apart intermediate configuration referred to above. This configuration is achieved and maintained by the cooperation of the free ends **28** of the branches **16a**, **16b** with the steps **40** of the female fitting **14**.

(62) However, the pin **16** can retain some a mobility in the inclined plane P2 in which it is located, by sliding on the surfaces **44b** of the steps **40**.

(63) In the operative position illustrated in FIG. **9** and following, the pin **16** is located in a plane perpendicular P3 to the axis X. In this position, the branches **16a**, **16b** of the pin **16** can adopt the two aforementioned extreme positions, spaced apart and close together, in the plane. P3

(64) In the operative position, the portions **31** of the branches **16a**, **16b** are located at the level of the slits **32** and are able to pass through them by elastic return. The branches **16a**, **16b** then move from a spaced apart position to the close together position shown in FIGS. **9** and **10**. FIG. **9** shows that in this position, the free ends **28** of the branches **16a**, **16b** are housed just above the stops **54**, and their portions **31** are housed in the gorge **22** of the male fitting **12**. The male fitting **12** is then blocked axially in the female fitting **14**.

(65) To remove the male fitting **12** from the female fitting **14**, simply remove the pin **16** from the gorge **22** in the male fitting **12**. To achieve this, a radial tensile force can be applied to the bridge **16c** of the pin **16** to move it apart from the female fitting **14**. The movement of the pin **16** is guided by the cooperation of the branches **16a**, **16b** with the lateral walls of the gorge **22** and is produced in the plane P3. During this movement, the free ends **28** of the branches **16a**, **16b** slide over the

surface portions **52** and become housed in the recesses **50** which thus maintain the branches **16a**, **16b** in the extreme spaced apart configuration shown in FIGS. **12a** and **12b**. The spaced apart position of the branches **16a**, **16b** could also be considered as illustrated in FIG. **9** insofar as the axial position of the pin **16** remains the same, and is further illustrated by dotted lines in FIG. **10**.

(66) According to the invention, the pin **16** is moved axially from its inoperative position to its operative position by means of the male connector **12**. In other words, it is the male fitting **12** itself which, when inserted into the female fitting **14**, causes the pin **16** to move axially. When the male fitting **12** is inserted, its frustoconical part **14a2** rests on the portions **31** of the branches **16a**, **16b**. When the insertion is continued, the branches **16a**, **16b** slide over the part **14a2** and move apart from each other, passing from the distance **D3** to the distance **D2**. The pin **16** is moved until it rests axially on the walls **32a** of the slits **32**. The pin **16** is then in the plane **P3**. When the axial position of the male fitting **12** in the female fitting **14** is such that the gorge **22** of the male fitting **12** is also located in the plane **P3**, the branches **16a**, **16b** are housed in the gorge **22** by elastic return.

(67) The greater **D3**, i.e. the further apart the branches **16a**, **16b** are spaced, the less force is required to engage the male fitting **12** with the female fitting **14**. **D3** is therefore preferably closer to **H2** than to **D1**. In practice, this means that the portions **31** of the branches **16a**, **16b** will bear on areas **Z** of the part **12a2** of the male fitting **12**, which are closer to its larger diameter end **H2** than to its smaller diameter end **H1**. These areas **Z** are illustrated schematically in FIGS. **2** and **3**.

(68) The present invention also relates to a fluid circuit for a motor vehicle, comprising at least one connector **10** as described above.

(69) The present invention also relates to a method for assembling the fluidic connector **10**, comprising the aforementioned steps of: mounting the pin **16** on the female fitting **14**, with the pin **16** in its inoperative position, and engaging the male fitting **12** in the female fitting **14**, this engagement causing the pin **16** to move from its inoperative position to its operative position in which said portions **31** are located at the level of said at least one slit **32** and are housed in the gorge **22** of the male fitting **12** by elastic return.

Claims

1. A fluidic connector, comprising: a female tubular fitting having a first axis, a tubular male fitting having a second axis and configured to engage coaxially with the female fitting, and a U-shaped metal pin mounted astride on the female fitting, the pin comprising two lateral branches, respective portions of the two lateral branches are configured to pass through at least one respective slit in the female fitting and to be engaged in a gorge in the male fitting to block the male fitting axially with respect to the female fitting, the pin being elastically deformable by moving the branches apart and close together and being able to adopt several configurations, between a close together configuration in which said portions are at a distance **D1** from one another and are housed in the gorge, and a spaced apart configuration in which said portions are at a distance **D2** from one another and allow the male fitting to be removed from the female fitting, **D2** being greater than **D1**, wherein the pin is axially movable on the female fitting from an inoperative position in which the branches are held spaced apart by the female fitting so that said portions are at a distance **D3** from one another, **D3** being between **D1** and **D2**, to an operative position in which said portions are located at a level of said at least one slit and are able to pass through the slit by elastic return, and wherein the male fitting comprises a part configured to be engaged in the female fitting and comprising an external diameter which is greater than **D3**, so that, when the male fitting is engaged in the female fitting, the part cooperates directly with said portions to move the pin from the inoperative position to the operative position.
2. The fluidic connector according to claim 1, wherein in said operative position, the pin extends substantially in a plane perpendicular to said first axis, and in said inoperative position the pin extends substantially inclined with respect to said first axis.

3. The fluidic connector according to claim 2, wherein, in said inoperative position, the pin is inclined so that free ends of the branches are located downstream from opposite ends of the branches, relative to a direction of engagement of the male fitting with the female fitting.
 4. The fluidic connector according to claim 1, wherein the female fitting comprises two of the at least one slit which are substantially diametrically opposed with respect to said first axis and which each have an elongate shape in a circumferential direction about said first axis, each of the slits being configured to be passed through by one of said portions.
 5. The fluidic connector according to claim 4, wherein the female fitting further comprises two notches which are substantially diametrically opposed with respect to said first axis and which communicate respectively with said slits, each of the notches having a lateral wall which is inclined with respect to a plane perpendicular to said first axis and on which one of said portions is adapted to engage by axial abutment when the pin is in the inoperative position.
 6. The fluidic connector as claimed in claim 5, wherein each of the notches is V-shaped or wedge-shaped and comprises a thin end and a widened end which is located on a side of a free end of the corresponding one of the branches.
 7. The fluidic connector according to claim 5, wherein each of the notches comprises a first part which opens into an interior of the female fitting, and a second part which defines a step, the step being configured to cooperate by radial abutment with a free end of the corresponding one of the branches to maintain the pin in the inoperative position.
 8. The fluidic connector according to claim 7, wherein said step comprises a projecting wedge on a side of the corresponding slit, the projecting wedge comprising a lug for retaining the pin in the inoperative position.
 9. The fluidic connector according to claim 8, wherein said step is connected, on the side opposite the lug, to a stop for the pin.
 10. The fluidic connector according to claim 7, wherein the steps of the female fitting comprise parallel surfaces for sliding the free ends of the branches.
 11. The fluidic connector according to claim 1, wherein, in the operative position, the pin is able to move in a plane perpendicular to said first axis, and in the inoperative position, the pin is able to move in a different plane.
 12. The fluidic connector according to claim 1, wherein said part of the male fitting is a frustoconical part which comprises an upstream end of smaller diameter and a downstream end of larger diameter, the part cooperating by bearing with said portions in areas which are closer to the downstream end than to the upstream end.
 13. The fluidic connector according to claim 1, wherein the female fitting comprises recesses which are substantially diametrically opposed with respect to said first axis, and which are configured to accommodate bent free ends of said branches when the pin is moved from the operative position to a disassembly position in which the pin has said spaced apart configuration.
 14. The fluidic connector according to claim 1, wherein, in the operative position, the pin is able to move in a plane perpendicular to said first axis, and in the inoperative position, the pin is able to move in a different plane which is inclined with respect to said first axis.
 15. A fluid circuit for a motor vehicle, comprising at least one connector according to claim 1.
 16. A method for assembling a fluidic connector, comprising the steps of: providing the fluid connector of claim 1; mounting the pin on the female fitting, with the pin in the inoperative position, and engaging the male fitting in the female fitting, the engaging causing the pin to move from the inoperative position to the operative position in which said portions are located at the level of said at least one slit and are housed in the gorge of the male fitting by elastic return.
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